MACCHANGER AND WAZUH OPEN SOURCE SIEM

Practical Project Report

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Environment	Kali Linux running in VMware Workstation
Project Scope	MAC address randomization and Wazuh Manager verification

Submitted as practical documentation for the cybersecurity project.

1. Project Overview

This project demonstrates two practical activities in a Kali Linux virtual machine: changing and verifying a network interface MAC address using Macchanger, and installing, starting, and checking the Wazuh Manager service.

> The work is focused on command-line practice and verification. The Wazuh Manager was running and its communication/API ports were observed listening. The separate Wazuh Indexer and Dashboard components were not deployed in this lab.

2. Objectives

- Understand the role of a MAC address in local network communication.
- Use Macchanger to display, randomize, verify, and restore a MAC address.
- Understand basic Linux service management using systemctl.
- Start and verify the Wazuh Manager service.
- Check relevant listening ports and interpret the observed output.

3. Lab Environment and Tools

Operating system: Kali Linux Rolling, running in VMware Workstation.
Network interface used: eth0.
Tools: Macchanger, iproute2 utilities, systemd/systemctl, Wazuh Manager, ss, curl, and ping.

4. Practical Part A — Macchanger

4.1 Identify the current and permanent MAC address

Command: `sudo macchanger -s eth0`
This displays the current MAC address and the permanent address associated with the interface. In the captured practical, the starting current and permanent MAC addresses matched.

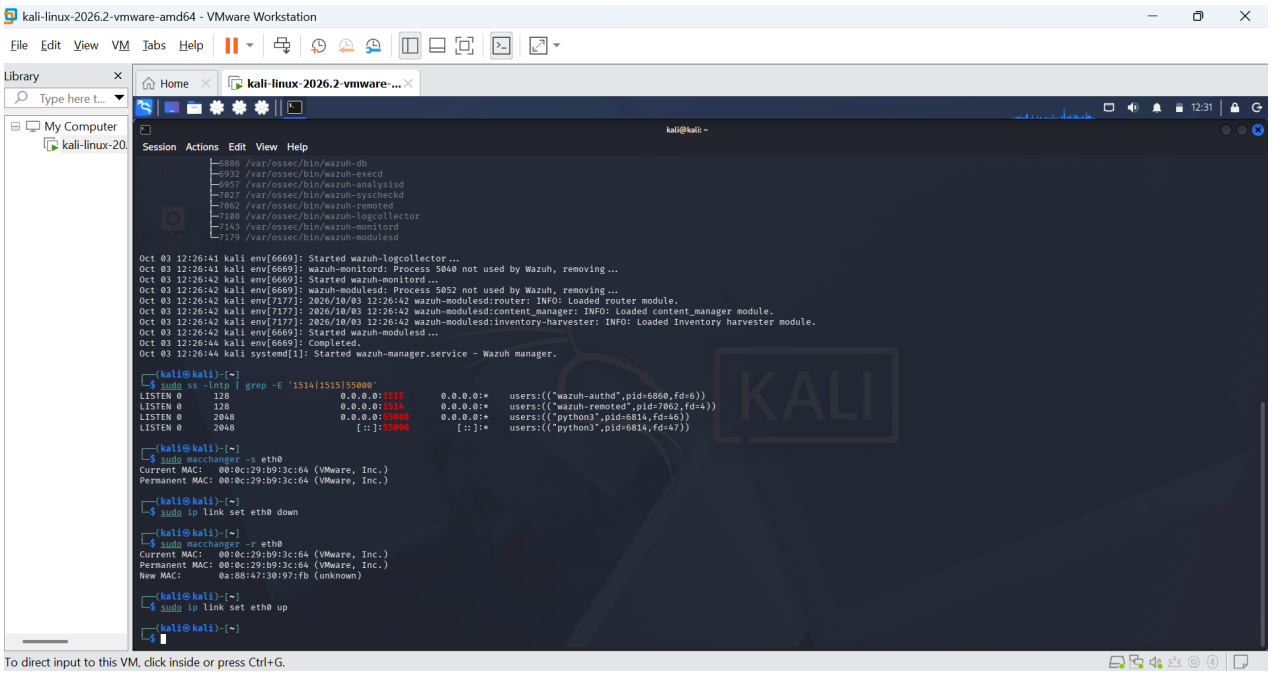


Figure 1. Terminal evidence showing the MAC address checks and randomization commands.

4.2 Temporarily disable the interface and randomize the MAC

Commands:

```
sudo ip link set eth0 down
```

```
sudo macchanger -r eth0
sudo ip link set eth0 up
```

The interface is taken down before the change, Macchanger generates a random address with the -r option, and the interface is brought up again. The captured output shows a randomized current MAC different from the permanent VMware MAC.

4.3 Verify and restore

The command `sudo macchanger -s eth0` was used to verify the randomized address. The permanent address was then restored with `sudo macchanger -p eth0` while the interface was down, followed by bringing the interface up. The final check showed the current MAC matching the permanent MAC.

5. Practical Part B — Wazuh Manager

Wazuh is an open-source security platform used for security monitoring and analysis. In this lab, the work focused on the Manager component. The Manager service was started and verified using systemd. This is a Manager-focused practical, not a complete multi-component deployment.

5.1 Check the service state

Command: `sudo systemctl status wazuh-manager --no-pager`
The command displays the service state without opening a pager. The captured terminal output showed **Active: active (running)**, confirming that systemd reported the Wazuh Manager service as running.

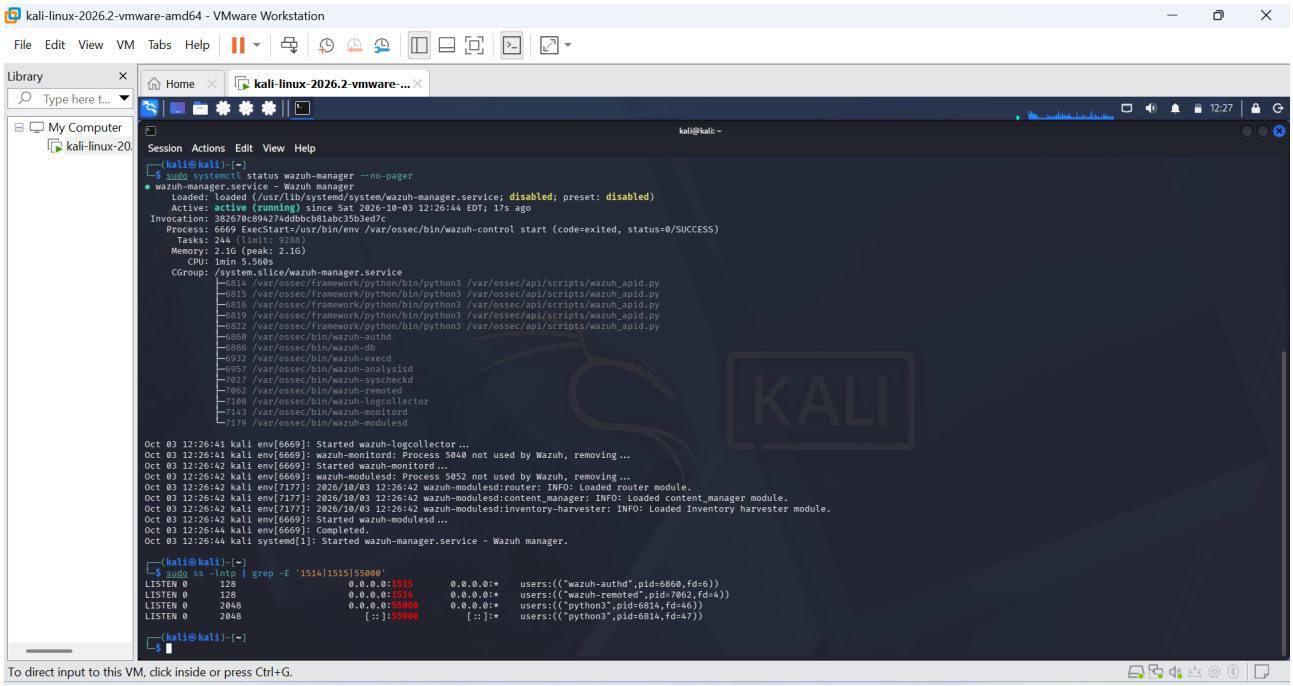


Figure 2. Wazuh Manager service status and listening-port verification in Kali Linux.

5.2 Verify listening ports

Command: `sudo ss -lntp | grep -E '1514|1515|55000'`
The ss utility displays socket information; the selected options show listening TCP sockets and process information. grep filters the output for the selected ports. The observed output showed port 1514, port 1515, and port 55000 listening. These correspond to Wazuh agent communication, agent enrollment, and the Wazuh API respectively.

5.3 API observation

A request to the local API endpoint confirmed that the API process was reachable on port 55000. An authentication attempt returned an Unauthorized/Invalid credentials response. Therefore, API authentication was not completed in this practical and no successful authenticated API operation is claimed.

6. Connectivity Observation

A ping test to google.com was performed after the MAC address change. The captured result showed 2 replies from 4 packets transmitted (50% packet loss). This is recorded as an observation only; it does not establish stable connectivity.

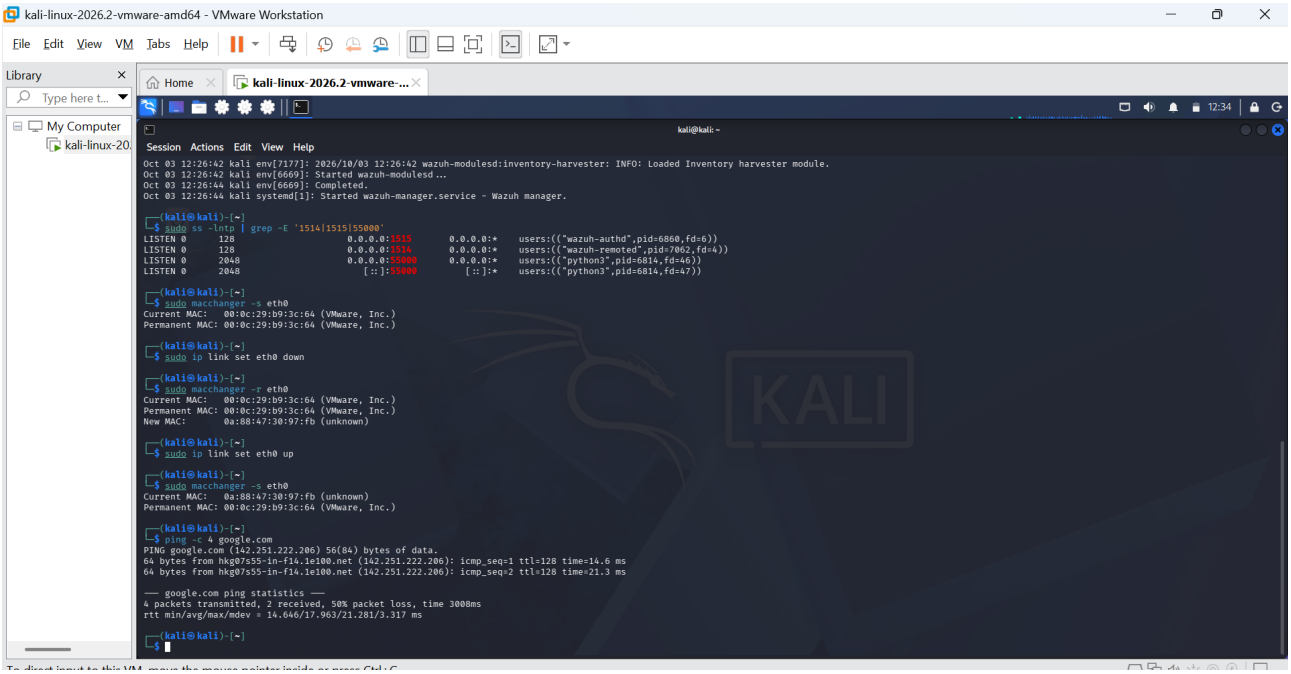


Figure 3. Captured connectivity test output.

7. Results and Learning

The Macchanger demonstration showed that a randomized MAC address could be assigned to eth0 and then restored to the permanent address. The Wazuh Manager service was active, and the selected Wazuh-related ports were listening at the time of verification.

The project provided practical experience with Linux terminal commands, network interface administration, service status checks, socket/port inspection, and introductory SIEM concepts.

8. Scope and Limitations

- The practical was performed in a Kali Linux virtual machine.
- The Wazuh Manager and API were checked; the Indexer and Dashboard were not installed/deployed.
- API authentication returned invalid credentials and was not resolved in the captured evidence.
- The ping test showed 50% packet loss, so stable connectivity is not claimed.
- MAC address randomization is not encryption and does not make a system fully anonymous.

9. Conclusion

This project completed a practical demonstration of MAC address randomization and restoration using Macchanger, along with installation and service-level verification of the Wazuh Manager. The work strengthened understanding of Linux networking commands, service management, and basic security monitoring architecture.

10. Safety and Ethical Use

All demonstrations were conducted in the user's own virtual lab environment for educational purposes. MAC address changes and security monitoring should only be performed on systems and networks where the user has authorization.

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